Final Project - Deploy Your Machine Learning models on hardware!

ESE 3600 : TinyML

Fall 2025

A. Project Guidelines

The aim of the final project is to demonstrate your proficiency in implementing the concepts learned in this course for a practical scenario. You have the freedom to incorporate ideas beyond the course syllabus along with the ones covered in the course.

Each project team will have **two** members. Use Ed Discussion Teammate Search if you need a partner.

Your final project is expected to have the following components at minimum:

1. *Extensive Dataset Collection and Preprocessing* - you must capture, curate, use and make freely available the dataset needed for your project.
2. *Machine Learning Model Analysis* - develop your model and show the analysis of why you used such a model. Feel free to use Transfer Learning and use models that are not limited to the ones covered in the assignments and tutorials.
3. *Extensive Model Evaluation* - evaluate your model in the context of your application to show how well it works, give examples of it working well and highlight cases where it does not do as well. This step must demonstrate that you have a sufficiently accurate model that has been trained on an adequate size dataset, has been evaluated with a validation and test set.
4. *Hardware Implementation* - all projects must have a hardware element. This could be with interfacing the robot platform with the Arduino or with the smartphone via bluetooth. Ideally, you will not have a software-only project running on the phone. We are keen to have you interface with discrete sensors and interface with the physical world.

Platforms:

You have the choice to pick the platform you want to deploy the final project model on. In this way, you will no longer be constricted by the small memory and processing capabilities of your hardware device.

1. Deploy on **Xiao** as seen in class
2. Deploy on your **Smartphone**! Steps to run your TF models on a mobile device - Link here



You will also be able to use this really fun and easy to drive mobile robot platform with your Xiao

2. Deliverables:

The final project has the following graded milestones:

1. **Project Proposal** due 11/17/2025. This should be a **two** page document with the 'title', 'introduction', 'motivation', and **'**platform' chosen. Explain what the problem is and how you will approach it with your TinyML solution. Include a timeline
2. **Intermediate Project Check-ins** due November 19th - 24th 2025. Start building a quick-n-dirty prototype, start collecting your data, sketch out what you expect the final output to look like. Teams will update and discuss with the staff about the project progress during Office Hours.
3. **Demo Day on** 12/01/2025. Show your working system or a short demo of it in action.
4. **Final Project Presentation** is on Friday 12/05/2025. Teams will give a **10 minute** presentation on their final projects to the class.
5. **Final Project Report** & **Code Submission** due 12/11/2025.
    a) Project Report - Teams need to submit a **6 page** document describing project details. The report should be structured under the following headings:
        i) Introduction
        ii) Motivation
        iii) Dataset Details
        iv) Model Choice/Design (eg: Why was a particular model chosen & corresponding model details)
        v) Model Training and Evaluation Results
        vi) Deployment/Hardware Details
        vii) Challenges/ Future Work
    b) Code Submission - Submit **Code** (colab notebook, arduino files, preprocessing scripts etc) and **Dataset** as a zip file.
    c) Video: Make a 90 sec-120 sec demo reel of your system. Think of this as a product pitch of the final working prototype. So start with the demo and then explain how it works (not the other way around).

3. Project Rubric:

The distribution of points for the Final Project is as follows:

Criteria	**Points (Total = 100)**
Project Proposal & Check-ins	10
Creativity and Uniqueness	10
Dataset Collection & Preprocessing	20
Results & Evaluation	20

Hardware Implementation	10
Project Presentation	10
Project Report, Video, and Code Submission	20

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